

Research Interests

- Birational geometry and the minimal model program: Sarkisov program, algebraically integrable foliations, and moduli spaces.
 - AI4Math: human verification of AI-generated mathematical results, Lean 4 formalization, training approaches for math-specialized models, and research-harness building.
 - Currently learning: PyTorch / libtorch, LLM pre-training and post-training, and mathematical data processing.
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Education

2024 - present **Ph.D. in Mathematics (in progress)**, AMSS, University of Chinese Academy of Sciences – expected June 2027.

Advisor: Prof. Yifei Chen.

Research: birational geometry – Sarkisov program, algebraically integrable foliations.

2022 - 2024 **M.S. in Mathematics**, AMSS, University of Chinese Academy of Sciences – May 2024.
Thesis: “Sarkisov program for foliated pairs”.

Advisor: Prof. Yifei Chen.

2016 - 2020 **B.S. in Mathematics**, Beihang University – June 2020.
Thesis: “Moduli space of curves”.

Awards and Honors

- MCM/ICM – Meritorious Winner (twice)
 - Hua Luogeng Scholarship, Beihang University (twice)
 - CAS – Merit Student of the Chinese Academy of Sciences
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Publications

(1) Sarkisov Program for Algebraically Integrable Adjoint Foliated Structures.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Mathematics Research Notices*, 2026(6):rnag045. 2026

Abstract: By applying the theory of the minimal model program for adjoint foliated structures, we establish the Sarkisov program for algebraically integrable foliations on klt varieties: any two Mori fiber spaces of such a structure are connected by a sequence of Sarkisov links. Combining with a result of R. Mascharak, we establish the Sarkisov program for foliations in dimension at most 3 with mild singularities. Log and adjoint-foliated versions of the aforementioned Sarkisov programs are also established.

(2) Flop between algebraically integrable foliations on potentially KLT varieties.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Journal of Mathematics*, 36(11):2550035. 2025

Abstract: We prove that for any two minimal models of an lc algebraically integrable foliated triple on potentially klt varieties, there exist small birational models that are connected by a sequence of flops. In particular, any two minimal models of lc algebraically integrable foliated triples on $\mathbb{Q}$ -factorial klt varieties are connected by a sequence of flops. We also discuss the connection between minimal models for possibly non-algebraically integrable foliations on threefolds, assuming the minimal model program for generalized foliated quadruples.

(3) A Note on the Sarkisov Program.

Yifei Chen and Yanze Wang. *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov, London Mathematical Society Lecture Note Series, pages 231–263, Cambridge University Press. 2025*

Abstract: This note introduces three methods of the Sarkisov program, which factorizes birational maps of log Mori fibre spaces into elementary Sarkisov links.

Preprints

(1) **A klt generalized pair with infinitely generated canonical ring** (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2608.03258. 2026*

Abstract: We construct a projective klt generalized pair (X, B, M) with infinitely generated generalized log canonical ring $R(X, K_X+B+M)$. This gives the first example of a klt variety X with $-K_X$ nef whose anticanonical ring $R(X, -K_X)$ is not finitely generated.

(2) **Twelve common flex lines in a general pencil of cubics** (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2607.26396. 2026*

Abstract: We prove that a general pencil of plane cubics over $\mathbb{C}$ has exactly 12 common flex lines. This answers a question of Ciliberto, Miranda, and Roe.

(3) **A counterexample to the odd-dimensional rank bound for abelian p-group actions** (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2607.04891. 2026*

Abstract: We give a counterexample to the odd-dimensional extension of the rank bound for faithful abelian p-group actions on smooth Calabi-Yau varieties, conjectured by Moraga.

Theses

- **M.S.:** “Sarkisov program for foliated pairs” (2024).

Abstract: the minimal model program seeks a good representative in each birational equivalence class; these representatives split into minimal models and Mori fiber spaces. The thesis introduces the three methods of the Sarkisov program and establishes it for foliated pairs – by reducing F-dlt foliated pairs to klt pairs, one obtains a weak decomposition of birational maps between foliated Mori fiber spaces.

- **B.S.:** “Moduli space of curves” (2020).

Abstract: this report constructs moduli spaces of curves and their compactifications via geometric invariant theory (GIT) and stack theory – introducing smooth and DM stable curves, developing the necessary tools (category theory, stack theory, Hilbert scheme, GIT), constructing two kinds of curve moduli spaces, and closing with further questions.

Invited Talks

2026 Speaker, **Sarkisov Program for Algebraically Integrable and Threefold Foliations**, Xi’an Jiaotong-Liverpool University.

Software and Open Source

Languages & tools: C++ (libtorch), Rust, C, Shell; daily development on Arch Linux (Niri/dwm)

+ Neovim; Git; LaTeX; TUI development (ratatui).

Calman Task & event manager with CLI and TUI (Rust). JSONL/ICS storage, CalDAV-compatible; syncs across devices with a self-hosted Radicale server by delegating to git / vdirsynchroner / rclone. Vibe-coded.

Markerss TUI RSS reader (Rust, ratatui + feed-rs): Markdown export, URL-only bookmarks. Design-first repo: main branch keeps SPEC / PLAN / DESIGN; implementations on per-language branches (rust / go / cpp). Vibe-coded.

Lichtung Hugo theme (Go templates / SCSS) built with conversational AI-assisted development; powers two personal sites (docs & blog).

suckless Personal fork of dwm / dwmblocks (X11 dynamic tiling window manager and related desktop components, written in C), patched and maintained by hand (no AI assistance).

AiKit AI tool collection for coding agents (especially pi): agents (pi, opencode, codex, claude code, ...), pi extensions (rtk, cave, toon, doc, role), skills, personas, and quick-deploy scripts.

Rethlas Forked and extended Rethlas: a natural-language math-reasoning system built around two Codex agents – a generation agent writes informal proof blueprints, a verification agent checks them and issues structured verdicts; run.sh drives an iterative proof-and-repair loop.

mumble A simple message-board page (JavaScript) on a Cloudflare Worker: deploy standalone or embed into other pages.

cutui A simple TUI video editor (Rust).

AI4Math Engineering

- **Research harness:** built and maintained an AI-assisted mathematics workflow (fork and extension of Rethlas).
- **Lean 4:** learning formalization and proof-repo maintenance.
- **Learning & exploring:** LLM pre-training, post-training, and mathematical data processing; PyTorch / libtorch applications in mathematics.